

SQL Interview Questions & Answers

A curated list of common SQL interview questions ranging from basic to advanced concepts.

1. What is SQL?

- SQL stands for Structured Query Language.
- It is the standard language used to communicate with Relational Database Management Systems (RDBMS) like MySQL, PostgreSQL, and SQL Server.
- We use it to create tables, store data, update data, and fetch data.

2. What are the different types of SQL commands?

- DDL (Data Definition Language): Defines structure (CREATE, ALTER, DROP, TRUNCATE).
- DML (Data Manipulation Language): Manages data (INSERT, UPDATE, DELETE).
- DCL (Data Control Language): Manages permissions (GRANT, REVOKE).
- TCL (Transaction Control Language): Manages transactions (COMMIT, ROLLBACK).

3. What is the difference between DELETE, TRUNCATE, and DROP?

- DELETE: Removes specific rows (can use WHERE). It is slower and can be rolled back.
- TRUNCATE: Removes ALL rows instantly. Resets auto-increment IDs. Cannot use WHERE.
- DROP: Deletes the entire table structure and data. The table ceases to exist.

4. What is a Primary Key?

- A column (or set of columns) that uniquely identifies every row in a table.
- It cannot contain NULL values.
- Example: `user\_id` or `employee\_id`.

5. What is a Foreign Key?

- A column that creates a link between two tables.
- It refers to the Primary Key of another table.
- It ensures "Referential Integrity" (you can't have an order for a user who doesn't exist).

6. What is the difference between WHERE and HAVING?

- WHERE: Filters rows BEFORE they are grouped. (e.g., "Find employees over age 30").

- HAVING: Filters groups AFTER the aggregation is done. (e.g., "Find departments where the average salary is > 50k").

7. Explain the different types of JOINS.

- INNER JOIN: Returns rows when there is a match in BOTH tables.
- LEFT JOIN: Returns all rows from the left table, and matched rows from the right table (NULL if no match).
- RIGHT JOIN: Returns all rows from the right table, and matched rows from the left table.
- FULL JOIN: Returns rows when there is a match in EITHER table.
- CROSS JOIN: Returns the Cartesian product (every row of table A combined with every row of table B).

8. What is an Index?

- An Index is a data structure that improves the speed of data retrieval.
- Think of it like the index at the back of a book. Instead of reading every page to find "Harry Potter", you look at the index to find the page number instantly.

Trade-off: Indexes speed up SELECT queries but slow down INSERT/UPDATE queries (because the index must be updated too).

9. What is a Transaction? (ACID Properties)

- A Transaction is a single unit of work (a group of tasks).
- It must follow ACID properties:
 - Atomicity: All or nothing. If one part fails, the whole transaction fails.
 - Consistency: The database goes from one valid state to another.
 - Isolation: Transactions don't interfere with each other.
 - Durability: Once committed, the changes are permanent (saved to disk).

10. What is the difference between RANK, DENSE_RANK, and ROW_NUMBER?

- ROW_NUMBER(): Assigns a unique number (1, 2, 3, 4). No duplicates.
- RANK(): Skips numbers for ties (1, 1, 3, 4).
- DENSE_RANK(): Does not skip numbers for ties (1, 1, 2, 3).

11. What is a View?

- A View is a virtual table based on the result-set of an SQL statement.
- It doesn't store data itself; it just saves a complex query so you can reuse it easily.

Example: `CREATE VIEW active_users AS SELECT * FROM users WHERE is_active = 1;`

12. What is Normalization?

- The process of organizing data to reduce redundancy (duplicates) and improve data integrity.
- 1NF: Atomic values (no lists in a single cell).
- 2NF: No partial dependency (all columns depend on the whole primary key).
- 3NF: No transitive dependency (columns shouldn't depend on other non-key columns).

13. What is the difference between CHAR and VARCHAR?

- CHAR: Fixed length. If you define CHAR(10) and store "Hi", it uses 10 bytes (pads with spaces). Faster for fixed-size data like Zip Codes.
- VARCHAR: Variable length. If you define VARCHAR(10) and store "Hi", it uses only 2 bytes (plus length info). Saves space.

14. What is a Stored Procedure?

- A prepared SQL code that you can save, so the code can be reused over and over again.
- You can pass parameters to it.
- Example: ``CALL AddUser('John', 'Doe');``

15. What is a Trigger?

- A set of SQL statements that automatically executes (or "fires") when a specific event occurs in the database.
- Events: INSERT, UPDATE, DELETE.
- Example: Automatically saving a log entry whenever a user is deleted.

16. What is NULL?

- NULL represents a missing or unknown value.
- It is NOT the same as zero (0) or an empty string ("").
- You must use ``IS NULL`` or ``IS NOT NULL`` to check for it (you cannot use ``= NULL``).

17. What is the difference between UNION and UNION ALL?

- UNION: Combines the result of two queries and REMOVES duplicates.
- UNION ALL: Combines the result of two queries and KEEPS duplicates. It is faster because it doesn't check for duplicates.

18. What is a Self Join?

- A regular join, but the table is joined with itself.

- Useful for hierarchical data (e.g., Employees and Managers, where the Manager is also an Employee in the same table).

19. What is a CTE (Common Table Expression)?

A temporary result set that you can reference within a SELECT, INSERT, UPDATE, or DELETE statement.

Defined using the `WITH` keyword.

- Makes complex queries easier to read.

20. How do you find the 2nd highest salary?

Method 1 (Limit/Offset): `SELECT salary FROM employees ORDER BY salary DESC LIMIT 1 OFFSET 1;`

Method 2 (Subquery): `SELECT MAX(salary) FROM employees WHERE salary < (SELECT MAX(salary) FROM employees);`

- Method 3 (Window Function):

```
WITH RankedSalaries AS (  
    SELECT salary, DENSE_RANK() OVER (ORDER BY salary DESC) as rank_val  
    FROM employees
```

-)

```
SELECT salary FROM RankedSalaries WHERE rank_val = 2;
```